

Birla Institute of Technology & Science - Pilani, Hyderabad Campus

Second Semester 2022-23

BITS F464 – Machine Learning

Comprehensive Examination

Type: Closed

Time: 180 mins

Max Marks: 90

Date: 14.05.2023

All parts of the same question should be answered together.

1. a. Do you agree with the following statement: “Decision Trees need not handle missing values explicitly.” Support your answer with the appropriate reasoning. [4 Marks]

1.b. Discuss a method to combat overfitting in decision trees. [4 Marks]

1.c How complex is to build and test a random forest as compared to a decision tree. Explain with all possible details. [4 Marks]

1.d. Consider the following set of training examples: [4 Marks]

Instance	Classification	a1	a2
1	+	T	T
2	+	T	T
3	-	T	F
4	+	F	F
5	-	F	T
6	-	F	T

What is the information gain of a2 relative to these training examples? Provide the equation for calculating the information gain as well as the intermediate results.

2. You want to train a neural network to drive a car. Your training data consists of grayscale 64×64 pixel images. The training labels include the human driver’s steering wheel angle in degrees and the human driver’s speed in miles per hour. Your neural network consists of an input layer with $64 \times 64 = 4,096$ units, a hidden layer with 2,048 units, and an output layer with 2 units (one for steering angle, one for speed). You use the ReLU activation function for the hidden units and no activation function for the outputs (or inputs). Calculate the number of parameters (weights) in this network. You can leave your answer as an expression but be sure to account for the bias terms at appropriate places. [4 Marks]

2.b. In class, we derived back propagation algorithm for regression problem using sigmoid as the activation function. If we change the activation function to tanh function, how does the derivation change? Provide all steps involved in the change. [10 Marks]

2.c. The overfitting problem is prevalent in Artificial Neural Networks as well. L2 regularization is one of the popular ways to handle regularization. Consider a L_2 regularized single layer neural network model that predicts continuous univariate target attribute $y = \sigma(z) \in \mathbb{R}$ where $z = wh + b$ and $h = \sigma(v)$ where $v = w'x + b'$, and σ is an activation function. To train, we use mean squared error from the targets $t \in \mathbb{R}$ with L_2 penalty on w, b' given by $L = (1/2) (y - t)^2 + w^2 + (b')^2$ [6 Marks]

Note: You may assume that there is only one feature x and univariate target attribute t for the above problem. $(x_1, t_1), (x_2, t_2), \dots, (x_N, t_N)$ are N training examples and y_1, y_2, y_N are the predicted target attributes of this dataset. Write the loss as a function of the parameters w, b and compute directly $\partial L / \partial w$, $\partial L / \partial b'$ using the backpropagation algorithm.

3. a. Consider a two-input neuron that multiplies its two inputs x_1 and x_2 to obtain the output o . Let L be the loss function that is computed at o . Suppose that you know that $\partial L / \partial o = 5$, $x_1 = 2$ and $x_2 = 3$. Compute the values $\partial L / \partial x_1$ and $\partial L / \partial x_2$.

Notation: $\partial L / \partial o$ refers to the partial derivative of L with respect to o .

[3 Marks]

3.b. Consider a neural network with three layers including an input layer. The first (input) layer has four inputs x_1, x_2, x_3 and x_4 . The second layer has six hidden units corresponding to all pairwise multiplications. The output node o simply adds the values in the six hidden units. Let L be the loss at the output node. Suppose that you know that $\partial L / \partial o = 2$, and $x_1 = 1, x_2 = 2, x_3 = 3$ and $x_4 = 4$. Compute $\partial L / \partial x_i$ for each $i = 1, 2, 3, 4$.

[6 Marks]

Notation: $\partial L / \partial o$ refers to the partial derivative of L with respect to o .

Some authors may call the above 3-layer network as 2-layer network also. The above network essentially means there is an input layer, hidden layer and an output layer.

4. Here we explore the maximal margin classifier on a toy data set.

(a) We are given $n = 7$ observations in $p = 2$ dimensions. For each observation, there is an associated class label.

Obs.	X_1	X_2	Y
1	3	4	Red
2	2	2	Red
3	4	4	Red
4	1	4	Red
5	2	1	Blue
6	4	3	Blue
7	4	1	Blue

Sketch the observations.

(b) Sketch the optimal separating hyperplane, and provide the equation for this hyperplane $w_0 + w_1 X_1 + w_2 X_2 = 0$

(c) Describe the classification rule for the maximal margin classifier. It should be something along the lines of "Classify to Red if $w_0 + w_1 X_1 + w_2 X_2 > 0$, and classify to Blue otherwise." Provide the values for w_0, w_1 , and w_2 .

(d) On your sketch, indicate the margin for the maximal margin hyperplane.

(e) Indicate the support vectors for the maximal margin classifier.

(f) Argue that a slight movement of the seventh observation would not affect the maximal margin hyperplane.

(g) Sketch a hyperplane that is not the optimal separating hyperplane, and provide the equation for this hyperplane.

(h) Draw an additional observation on the plot so that the two classes are no longer separable by a hyperplane.

[12 Marks]

5. We consider here linear and non-linear support vector machines (SVM) of the form:

$$\begin{aligned} \min w_1^2/2 \quad \text{subject to} \quad & y_i(w_1 x_i + w_0) - 1 \geq 0, \quad i = 1, \dots, n, \quad \text{or} \\ \min \mathbf{w}^T \mathbf{w} / 2 \quad \text{subject to} \quad & y_i(\mathbf{w}^T \Phi_i + w_0) - 1 \geq 0, \quad i = 1, \dots, n \end{aligned}$$

where Φ_i is a feature vector constructed from the corresponding real valued input x_i . We wish to compare the simple linear SVM classifier ($w_1 x + w_0$) and the non-linear classifier ($\mathbf{w}^T \Phi + w_0$), where $\Phi = [x, x^2]^T$.

a. You are given three training points (1, +), (2, -) and (3, +). Are these three points linearly separable in the original feature space. You apply transformation function, Φ , on these training examples. Are these data points with transformed features linearly separable? If so, what is/are the decision boundary, support vectors and the width of the margin.

b. Consider two labelled points $(1, +)$ and $(3, -)$. Is the margin we attain using feature vectors where $\phi = [x, x^2]^T$. You apply transformation function, ϕ , on these training examples. Are these data points with transformed features linearly separable? If so, what is the decision boundary, support vectors and the width of the margin. Find out whether the margin of SVM for this problem (b) is *greater than/less than/equal to* the margin obtained in problem (a).

c. In general, for three training points $(1, +)$, $(2, -)$ and $(3, +)$, find out whether the margin we would attain using scaled feature vectors $\phi = [2x, 2x^2]^T$ is *greater than/less than/equal to* the margin obtained in part (a). [10 Marks]

6.a. Can we would expect the support vectors to remain the same in general as we move from a linear kernel to higher order polynomial kernels. Justify your answer. [3 Marks]

6.b. Do you agree with the statement: “In training a logistic regression model by maximizing the likelihood of the labels given the inputs we have multiple locally optimal solutions”? Justify your answer. [3 Marks]

6.c. Suppose there is a dataset such that Class 1 has 5 samples $c1 = \{(1,2), (2,3), (3,3), (4,5), (5,5)\}$ and Class 2 has 6 samples $c2 = \{(1,0), (2,1), (3,1), (3,2), (5,3), (6,5)\}$. The mean $m1$ of class C1 and $m2$ of class C2 are $(3, 3.6)$ and $(3.3, 2)$. The total within-class covariance matrix, S_w , is given by

$$\begin{bmatrix} 27.3 & 24 \\ 24 & 23.2 \end{bmatrix}$$

Using the Fisher Discriminant Analysis find the vector on to which the training vectors collapse to. Find out the one-dimensional threshold approximately to decide whether a given testing point belongs to Class 1 or Class 2 and thus determine the class of the two testing points $(3,5)$ and $(-2,3)$. [10 Marks]

7.a. Do you agree with the statement: “If we train a Naïve Bayes Classifier using infinite training data that satisfies all of its modeling assumptions (e.g., conditional independence), then it will achieve **zero training error** over these training examples”. Justify your answer [2 Marks]

7.b. Do you agree with the statement: “If we train a Naïve Bayes Classifier using infinite training data that satisfies all of its modeling assumptions (e.g., conditional independence), then it will achieve **zero true error** over test examples drawn from this same distribution”. Justify your answer. [2 Marks]

7.c. Suppose that you have a linear SVM binary classifier. Consider a point that is currently classified correctly, and is far away from the decision boundary. If you remove the point from the training set, and re-train the classifier, will the decision boundary change or stay the same? Explain your answer in one sentence. [3 Marks]